



Department of Economics

Problem Set 2: Panel Fixed Effects and Difference-in-Differences Asymptotics

Due: Friday, October 24 at 17:00 • Total Score: 100 Points

Problem 1: Two-Way Fixed Effects Demeaning and Algebraic Equivalence (50 Points)

Consider the standard panel regression model with unit fixed effects α_i and time fixed effects λ_t :

$$Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

- Define the within-group double demeaning transformation $\check{Y}_{it}$ and prove that $\check{\alpha}_i = 0$ and $\check{\lambda}_t = 0$.
- Derive the OLS estimator of δ applied to the double-demeaned equation.

Detailed Solution & Mathematical Derivation

Part (a): Define $\bar{Y}_i = \frac{1}{T} \sum_{t=1}^T Y_{it}$, $\bar{Y}_t = \frac{1}{N} \sum_{i=1}^N Y_{it}$, and $\bar{Y} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T Y_{it}$. The double-demeaned variable is:

$$\check{Y}_{it} = Y_{it} - \bar{Y}_i - \bar{Y}_t + \bar{Y}$$

For the unit effect α_i : $\bar{\alpha}_i = \alpha_i$, $\bar{\alpha}_t = \frac{1}{N} \sum_i \alpha_i = \bar{\alpha}$, and $\bar{\alpha} = \bar{\alpha}$. Hence:

$$\check{\alpha}_i = \alpha_i - \alpha_i - \bar{\alpha} + \bar{\alpha} = 0$$

Similarly, $\check{\lambda}_t = \lambda_t - \bar{\lambda} - \lambda_t + \bar{\lambda} = 0$.

Part (b): The transformed model is $\check{Y}_{it} = \delta \check{D}_{it} + \check{\varepsilon}_{it}$. The within estimator is:

$$\hat{\delta}_{FE} = \frac{\sum_{i=1}^N \sum_{t=1}^T \check{D}_{it} \check{Y}_{it}}{\sum_{i=1}^N \sum_{t=1}^T \check{D}_{it}^2} \quad \blacksquare$$

Problem 2: Equivalence of 2×2 DiD and Sample Means (50 Points)

In a 2-period, 2-group setting ($t \in \{1, 2\}$, $D \in \{0, 1\}$), where treatment occurs at $t = 2$ for the treated group:

- State the regression specification $Y_{it} = \beta_0 + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \delta (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it}$.
- Show that the OLS estimate $\hat{\delta}$ equals $(\bar{Y}_{T,2} - \bar{Y}_{T,1}) - (\bar{Y}_{C,2} - \bar{Y}_{C,1})$.

Detailed Solution & Mathematical Derivation

Evaluating the conditional expectations of Y for the four cells:

$$\mathbb{E}[Y \mid \text{Treat} = 0, \text{Post} = 0] = \beta_0 = \bar{Y}_{C,1}$$

$$\mathbb{E}[Y \mid \text{Treat} = 0, \text{Post} = 1] = \beta_0 + \beta_2 = \bar{Y}_{C,2} \implies \beta_2 = \bar{Y}_{C,2} - \bar{Y}_{C,1}$$

$$\mathbb{E}[Y \mid \text{Treat} = 1, \text{Post} = 0] = \beta_0 + \beta_1 = \bar{Y}_{T,1} \implies \beta_1 = \bar{Y}_{T,1} - \bar{Y}_{C,1}$$

$$\mathbb{E}[Y \mid \text{Treat} = 1, \text{Post} = 1] = \beta_0 + \beta_1 + \beta_2 + \delta = \bar{Y}_{T,2}$$

Solving for δ :

$$\delta = \bar{Y}_{T,2} - (\beta_0 + \beta_1 + \beta_2) = (\bar{Y}_{T,2} - \bar{Y}_{T,1}) - (\bar{Y}_{C,2} - \bar{Y}_{C,1}) \quad \blacksquare$$